

TAKE-HOME EXERCISE

13F Positioning Factor & Backtester

You have 48 hours from when you open this document. You may use any tools you like, including Claude Code, Cursor, ChatGPT, or anything else. We assume you will.

The Task

Build a research pipeline that:

1. Ingests 13F-HR filings from the SEC's EDGAR database for a universe of institutional holders you define.
2. Constructs one or more positioning factors from the holdings data. You define what "positioning" means — crowding, concentration shifts, smart-money following, ownership momentum, or something else entirely. We want to see your hypothesis and your reasoning.
3. Backtests the factor(s) against equity returns over a historical window you choose.
4. Presents results in self-contained file together with supporting text (if needed).

You define the scope. Which companies filing, which equities, which time window, which factor construction methodology — all your call. We expect you to make deliberate cuts in the range of possible options for all of these and defend them. Trying to do everything is a failure mode.

What We Want Delivered

1. A working pipeline, runnable from a clean clone with documented setup steps. "Working" means we can run it and see real 13F data flow through to factor construction and backtest output.

2. A memo, three to five pages, covering: your factor hypothesis — what signals you're extracting from 13F data and why you believe it should have predictive power; scope decisions and what you deliberately excluded; methodology for data ingestion, factor construction, and backtesting, with particular attention to point-in-time discipline (how you handle the reporting lag, rebalancing timing, and lookahead avoidance); backtest results with an honest assessment of what worked, what didn't, where you have low confidence, and what you think is noise vs. signal; the three things you would build next, in priority order, with reasoning.

How We Will Evaluate

- Quality of the factor hypothesis. Is there a coherent story for why this signal should carry information? Did you consider alternatives?
- Point-in-time discipline. Does your backtester correctly handle the 13F reporting lag (filings are due 45 days after quarter-end but often arrive later)? Is there any lookahead bias?

- Data engineering care. CUSIP mapping, filer deduplication, handling of amended filings (13F-HR/A), treatment of confidential treatment requests — these are where sloppiness shows.
- Backtest methodology. Transaction cost assumptions, rebalancing frequency, benchmark choice, how you handle the quarterly signal in a daily or monthly return framework.
- Your ability to defend every choice in a follow-up technical session. You should be able to explain or extend any part of your own system.

Submit a link to a private repo when you are done. We will schedule a 60-minute technical defense within 48 hours of submission.